Introduction to NLP, Exercise 1

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Discussion: Oct 29, 2026

- **Deadline:** The submission needs to be done via VC until the end of the day mentioned on the PDF. A late submission is not possible.
- **Groups:** You are allowed to work in groups. A group can consist of maximally three members. The members can change in each submission. The names that you put on the PDF are what we consider as the group.
- **Grading:** We plan to offer 11 exercise sheets. For each sheet, you can achieve 20 points. From the 220 points altogether, we will consider 176 points to be a perfect participation (if, for some reason, we offer fewer assignments, this number of required points will be reduced proportionally). The assignment submissions will contribute up to 20 % of the grade as bonus points.
- **Submission:** Only submit one PDF file. Program code should be part of that PDF file. Put all names and email addresses of your group members on the first page of this PDF file. Please do submit the PDF only once (not multiple times by multiple group members).
- **Sources:** We recommend to not use any "AI" or large language model for creating the solution to any subtask/task as a whole. You can use other sources (documentation, papers, stackexchange, ...). If you decide to use an "AI" or LLM (like ChatGPT, Hawki) for a part of your solution, provide the information which system you used, provide the complete prompt and the complete answer by the automatic system (in German or English) in an appendix of your submission. I need to be able to understand what your own contribution on top of the automatic result was. You only receive points for your own contribution on top of what you got from the source you provide. If you use an "AI" and do not provide the required reference or do not make that use transparent, we will consider this plagiarism.
- **Plagiarism:** When two groups submit solutions that are copied from each other or from a third source that is not made transparent in the submission, all these groups do not receive any points. The same holds if one group copies from a source that is no provided in the submission. The whole submission will be considered invalid, not only the task in which we detected the plagiarism. No points will be given for the whole submission. These sanctions do not exclude further measures.

Do you have questions? Send me an email (sabine.weber@uni-bamberg.de) or use the forum.

Task 1 (10 points)

In the previous class you have been introduced to speech processing using wave forms and spectrograms. You will now generate and examine these yourself.

- Download and install the programm Praat. You can find a free version here: `https://praat.org/`
- Record yourself saying the sentence provided below.

  "Peter put the blue book beside the big window."

  Examine the waveform. Where can you detect word boundaries or vowel sounds? Can you transcribe the sentence based on the waveform alone? Document your transcription and observations in your report with a screenshot of the waveform.
- Create a spectrogram of your recording. Try the following different window lengths: 5 ms, 10 ms, and 25 ms. Describe how the appearance of the spectrogram changes with different window lengths. Which setting is best for speech analysis and why?
- Look at the spectrogram and identify the vowels in the sentence. Locate their formants F1 and F2. Mark the formant patterns and the vowels they correspond to.
- Look at the entire spectrogram and try to transcribe the sentence using the International Phonetic Alphabet (IPA). Do repeated instances of the same sound look similar? Why or why not? What sounds are easy to identify and why? Include screenshots of the spectrograms and explain your observations.

Next, think more about the applications of speech processing in real-world scenarios. Watch the following sketch about a speech recognition failure: `https://www.youtube.com/watch?v=MNuFcIRlwdc`

- Why do you think the speech recognition system fails? Think about how different phonemes play a role here, and which phonemes might be the reason for the failure. What strategies do the characters in the sketch apply to solve the problem and why do they fail?
- How would you design an experiment to test your hypothesis why the system fails?
- How would you try to improve the system?

Task 2 - Programming (10 points)

There are dictionaries that map written words to their pronunciation. In the programming task you will use such a dictionary to look up the pronunciation of words to build a rhyme checker.

- Look up the CMU Prounouncing Dictionary. How does the systems used there differ from the IPA?
- Load the dictionary via the `pronouncing` library in Python.

- Use the dictionary to look up the pronunciation of the words in the following sentence to get a feel for how the dictionary represents pronunciation: "Peter put the blue book beside the big window."
- Write a function `are_rhymes(word1, word2)` that returns `True` for slant or exact rhymes (e.g., "skate" and "eight") despite mismatched spellings. Extract the nucleus and coda of the primary stressed syllable to define a valid rhyme condition.
- Test your function using the poem "The Chaos" by Gerard Nolst Trenité.